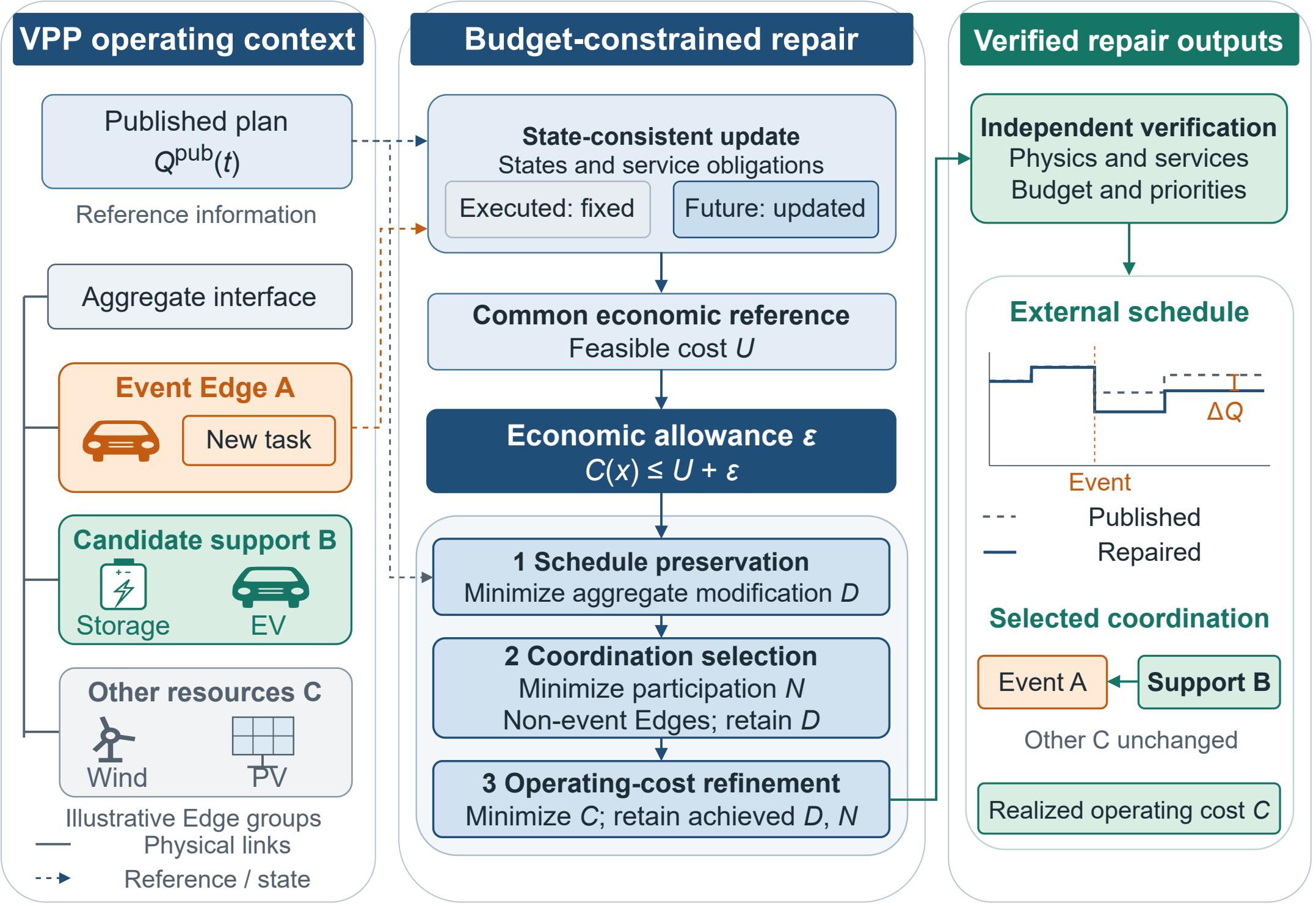

VPP operating context
Published plan
$Q^{pub}(t)$
Reference information
Aggregate interface
Event Edge A
New task
Candidate support B
Storage
EV
Other resources C
Wind
PV
Illustrative Edge groups
Physical links
Reference / state
Budget-constrained repair
State-consistent update
States and service obligations
Executed: fixed
Future: updated
Common economic reference
Feasible cost U
Economic allowance ε
$C(x) \leq U + \varepsilon$
1 Schedule preservation
Minimize aggregate modification D
2 Coordination selection
Minimize participation N
Non-event Edges; retain D
3 Operating-cost refinement
Minimize C; retain achieved D, N
Verified repair outputs
Independent verification
Physics and services
Budget and priorities
External schedule
ΔQ
Event
Published
Repaired
Selected coordination
Event A
Support B
Other C unchanged
Realized operating cost C